MPSTME LIBRARY
NARSEE MONJEE INSTITUTE OF MANAGEMENT STUDIES ★ NMIMS UNIVERSITY ★

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: **2025-26**

Program: **B TECH/MBA TECH/BTI(DS/IT/COMP/AI/AI&DS/AI&ML/COMP SCI/CSBS/CSE CS/CSE DS/EXTC/MXTC)**

Subject: **Data Structures and Algorithms** Year: **II/III/IV** Semester : **III/V/VII**

Date : **26 Nov 2025** Time **10.00 am to 01.00 pm (3 hrs)**

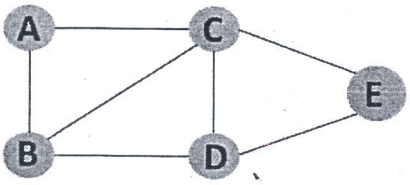
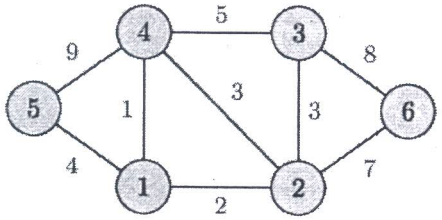
Marks: **100** No. of Pages: 2

Final Exam (2025-26) / Re-Exam (2023-24/2024-25)

Instructions:

1. This question paper contains 2 pages.
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Question No. 1 is compulsory.
5. Out of remaining questions, attempt any 4 questions

Q1		Answer briefly:	[20]
CO-2 ; SO-1,2 ; BL-3	a.	Define Stack. Consider the following scenario, stack is implemented using an array with a maximum size of 5. Perform the following operations and show the state of the stack after each step. Also, identify the outputs of operations like pop() and peek(). Operations to Perform: Push(10), Push(20), Push(30), Peek(), Pop(), Push(40), Push(50), Push(60), isEmpty(), Push(70)	
CO-4 ; SO-1,2 ; BL-3	b.	Show all intermediate passes using selection and insertion sort for the given set of elements: 4 3 1 5 2	
CO-3 ; SO-1,2 ; BL-3	c.	Create a binary search tree with the given elements: 98, 2, 48, 12, 56, 32, 4, 67, 23, 87, 23, 55, 46.	
CO-1 ; SO-1,2 ; BL-4	d.	Justify which data structure is most suitable for the given scenario- 1. A system needs to store passenger records with unique ticket IDs. The system must allow constant-time lookup to check whether a passenger exists. 2. An operating system must schedule processes such that the process with minimum running time is scheduled first.	
Q2 CO-1; SO-1,2; BL-3	a.	Explain time and space complexity. Explain various asymptotic notations used to calculate time complexity.	[10]
CO-2; SO-1,2; BL-3	b.	Write an algorithm for parenthesis checker using stack? Solve given expression for parenthesis check using stack: (a-c)*(b+c]	[10]
Q3 CO-2; SO-1,2; BL-3	a.	Differentiate between linked list and arrays. Write an algorithm to insert a new node after the given node in singly linked list.	[10]
CO-3; SO-1,2; BL-3	b.	Explain how deletion is performed in a Binary Search Tree (BST). Write an algorithm to delete a node with a given key from a Binary Search Tree.	[10]

Q4 CO-4; SO-1,2; BL-3	a.	Write algorithm for performing depth-first search on a graph data structure. Perform a DFS traversal starting from vertex A. Show all intermediate steps and write the final DFS order. 	[10]
CO-4; SO-1,2; BL-3	b.	Write algorithm for performing Merge sort. Show how the following set of elements will be sorted using Merge sort: 45 21 67 30 19 55 20 70.	[10]
Q5 CO-2; SO-1,2; BL-3	a.	Write algorithm to convert Infix notation to Postfix notation using stack. Apply the same on following expression: (A – 2 * (B + C) / D * E) + F	[10]
CO-3; SO-1,2; BL-3	b.	Write an algorithm to insert a new node into a binary search tree. Construct binary search tree from the given traversal: Preorder: 40, 20, 10, 30, 60, 50, 70 Postorder: 10, 30, 20, 50, 70, 60, 40	[10]
Q6 CO-3; SO-1,2; BL-3	a.	What is minimum spanning tree. Construct a minimum spanning tree of the given graph using Prim's algorithm. Start from vertex 5. Show all intermediate steps. 	[10]
CO-2; SO-1,2; BL-3	b.	Explain a doubly linked list and describe how it differs from a singly linked list. Write an algorithm to delete a node with a given value from a singly linked list. Consider both cases: when the node is found and when it is not found.	[10] ~~[05]~~
Q7 CO-2; SO-1,2; BL-3	a.	Explain the concept of a Circular Queue, highlighting its need over a linear queue. Write the algorithm for its primary operations: enqueue() and dequeue(), along with the conditions for queue overflow and underflow. Give any two real-world applications of circular queue.	[10]
CO-3; SO-1,2; BL-3	b.	Write Huffman encoding algorithm. A text file contains the following letter frequencies: P:30, Q:3, R:6, S: 12, T:34, U:7 Construct the Huffman tree and generate the Huffman codes. Show all intermediate steps.	[10]

28/11/2025